

Solution Guidelines

Problem 1. Let $\{a_n\}_{n \in \mathbb{N}}$ be a convergent sequence in $\mathfrak{R}$, let $\sigma : \mathbb{N} \rightarrow \mathbb{N}$ be **injective** and put $b_n = a_{\sigma(n)}$. Show that $\{b_n\}$ converges and $\lim b_n = \lim a_n$.

Why this is *not* just “a subsequence converges” σ need not be increasing: $\{b_n\}$ may reshuffle the terms (e.g. $\sigma = (2, 1, 4, 3, 6, 5, \dots)$), so it need not be a subsequence.

Key idea Fix $\varepsilon > 0$. Convergence gives an N such that $|a_m - a| < \varepsilon$ for every $m \geq N$. So among the indices of the original sequence, the only ones for which the term a_m might fail to be within ε of a are $m \in \{1, 2, \dots, N-1\}$. These are the “bad” indices, and there are only $N-1$ of them. Everything from N onwards is “good”.

Now, $b_n = a_{\sigma(n)}$ is within ε of a as soon as $\sigma(n)$ is a good index. So the question is: for how many n can $\sigma(n)$ be bad? Let

$$F = \sigma^{-1}(\{1, \dots, N-1\}) = \{n \in \mathbb{N} : \sigma(n) \leq N-1\}.$$

Because σ is injective, distinct $n \in F$ have distinct values $\sigma(n)$, and all those values lie in the $(N-1)$ -element set $\{1, \dots, N-1\}$. Hence $|F| \leq N-1$. Each bad index $m \leq N-1$ can be the value $\sigma(n)$ for at most one n , so at most $N-1$ inputs n get sent into the bad zone.

Since F is finite, it has a largest element (or is empty); put $M = 1 + \max F$ (and $M = 1$ if $F = \emptyset$). For $n \geq M$ we have $n \notin F$, i.e. $\sigma(n) \geq N$ - past the finitely many exceptional inputs, σ only takes good values. Therefore $|b_n - a| = |a_{\sigma(n)} - a| < \varepsilon$ for all $n \geq M$, and since ε was arbitrary, $b_n \rightarrow a$.

A picture that may help: an injective σ can shuffle the sequence arbitrarily, but it cannot *repeat* terms. The finitely many terms $a_1, \dots, a_{N-1}$ that are far from a can therefore appear at most once each in $\{b_n\}$, so they can only occupy finitely many positions; after the last of those positions, every b_n is one of the good terms.

In the proof, $F = \sigma^{-1}(\{1, 2, \dots, N-1\})$, i.e. the set of indices n whose image $\sigma(n)$ lands in the “bad” range $\{1, \dots, N-1\}$. Injectivity of σ gives an upper bound only: σ restricted to F is an injection into a set with $N-1$ elements, so $|F| \leq N-1$. Equality would need σ to be surjective onto $\{1, \dots, N-1\}$ as well - i.e. every one of the values $1, \dots, N-1$ actually gets hit by σ - and that is not assumed.

Concretely, take $\sigma(n) = 2n$. With $N = 4$ the bad range is $\{1, 2, 3\}$, and $F = \sigma^{-1}(\{1, 2, 3\}) = \{1\}$ (only $\sigma(1) = 2$ lands there), so $|F| = 1 < 3 = N-1$. At the other extreme, σ could miss the bad range entirely (e.g. $\sigma(n) = n + N$), giving $F = \emptyset$.

Proof. Let $a = \lim a_n$ and let $\varepsilon > 0$ be given.

1. There is $N \in \mathbb{N}$ with $|a_m - a| < \varepsilon$ for all $m \geq N$.
2. Consider the “bad” index set $F = \sigma^{-1}(\{1, 2, \dots, N-1\}) = \{n \in \mathbb{N} : \sigma(n) \leq N-1\}$. Since σ is injective, σ maps distinct elements of F to distinct elements of $\{1, \dots, N-1\}$, hence

$$|F| \leq N-1 < \infty.$$

3. Put $M = 1 + \max F$ (and $M = 1$ if $F = \emptyset$). If $n \geq M$ then $n \notin F$, i.e. $\sigma(n) \geq N$.
4. Therefore, for every $n \geq M$,

$$|b_n - a| = |a_{\sigma(n)} - a| < \varepsilon,$$

because $\sigma(n) \geq N$.

As $\varepsilon > 0$ was arbitrary, $b_n \rightarrow a = \lim a_n$. □

Two remarks

- **Injectivity is essential.** Take $a_n = 1/n$ and the constant map $\sigma \equiv 1$. Then $b_n = a_1 = 1$ for all n , so $b_n \rightarrow 1 \neq 0 = \lim_n a_n$.
- **Special case.** If σ is strictly increasing, $\{b_n\}$ is a *subsequence*; Q4 shows $\sigma(n) \geq n$, which is exactly step 3 above with $M = N$. So “every subsequence of a convergent sequence converges to the same limit” is a corollary.

Problem 2. State whether the following statements are True (T) or False (F). Give Reason

(a) Statement and Answer

Claim: For sequences $\{x_n\}, \{y_n\}$ in $\mathfrak{R}$: if $x_n \rightarrow x$ and $d(x_n, y_n) \rightarrow 0$, then $y_n \rightarrow x$.

The Statement is TRUE.

Idea: y_n is close to x_n , and x_n is close to x ; add the two errors with the triangle inequality.

Proof. Let $\varepsilon > 0$ be given.

- Since $x_n \rightarrow x$, there is N_1 with $|x_n - x| < \varepsilon/2$ for $n \geq N_1$.
- Since $d(x_n, y_n) = |x_n - y_n| \rightarrow 0$, there is N_2 with $|x_n - y_n| < \varepsilon/2$ for $n \geq N_2$.
- Let $N = \max\{N_1, N_2\}$. For $n \geq N$, the triangle inequality gives

$$|y_n - x| \leq |y_n - x_n| + |x_n - x| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence $y_n \rightarrow x$. □

Remark. Nothing was used about $\mathfrak{R}$ beyond the triangle inequality, so the same proof works verbatim in $\mathfrak{R}^n$ with $\|\cdot\|$.

(b) Statement and Answer

Claim: Let $\emptyset \neq S \subsetneq \mathfrak{R}$ be finite and $d(x, S) = \min_{z \in S} |x - z|$. Then for all $x_1, x_2 \in \mathfrak{R}$,

$$|d(x_1, S) - d(x_2, S)| \leq |x_1 - x_2|.$$

The Statement is TRUE

Preliminary: S is finite and non-empty, so the minimum is **attained**: for each x there is $z_x \in S$ with $d(x, S) = |x - z_x|$.

Proof. Fix $x_1, x_2 \in \mathfrak{R}$ and pick $z_2 \in S$ with $d(x_2, S) = |x_2 - z_2|$.

- Since $z_2 \in S$, the minimum defining $d(x_1, S)$ is at most the value at z_2 :

$$d(x_1, S) \leq |x_1 - z_2| \leq |x_1 - x_2| + |x_2 - z_2| = |x_1 - x_2| + d(x_2, S).$$

- Hence $d(x_1, S) - d(x_2, S) \leq |x_1 - x_2|$.
- Exchanging the roles of x_1 and x_2 gives $d(x_2, S) - d(x_1, S) \leq |x_2 - x_1| = |x_1 - x_2|$.
- Both differences are $\leq |x_1 - x_2|$, so $|d(x_1, S) - d(x_2, S)| \leq |x_1 - x_2|$.

□

Remark. Finiteness of S was used only to *attain* the minimum. For an arbitrary non-empty S , replace min by inf and run the same argument.

Problem 3. Let $\{a_n\}$ be real with $a_{2k} \rightarrow a_0$ and $a_{2k-1} \rightarrow a_0$ (as $k \rightarrow \infty$). Show $a_n \rightarrow a_0$.

Idea: Every index is even or odd. Get a threshold for each parity (odd or even) and take the larger one — **after translating the two thresholds from k -language to n -language**.

Warning: “ $k \geq K$ ” for the subsequence a_{2k} means “ $n = 2k \geq 2K$ ” for the original sequence. Forgetting this factor 2 is the standard slip in this exercise.

Proof. Let $\varepsilon > 0$ be given.

1. There is K_1 with $|a_{2k} - a_0| < \varepsilon$ for all $k \geq K_1$.
2. There is K_2 with $|a_{2k-1} - a_0| < \varepsilon$ for all $k \geq K_2$.
3. Put $N = \max\{2K_1, 2K_2 - 1\}$ and take any $n \geq N$.
 - If n is even, write $n = 2k$ with $k = n/2 \geq 2K_1/2 = K_1$, so $|a_n - a_0| < \varepsilon$ by (1).
 - If n is odd, write $n = 2k - 1$ with $k = (n + 1)/2 \geq (2K_2 - 1 + 1)/2 = K_2$, so $|a_n - a_0| < \varepsilon$ by (2).

In both cases $|a_n - a_0| < \varepsilon$ for all $n \geq N$; hence $a_n \rightarrow a_0$.

□

Problem 4. Let $n_1 < n_2 < n_3 < \dots$ be strictly increasing natural numbers. Show: for every $N \in \mathbb{N}$ there is k with $n_j > N$ for all $j \geq k$.

Lemma 1. $n_k \geq k$ for every $k \in \mathbb{N}$.

Proof. Induction on k .

Base: $n_1 \geq 1$ because $n_1 \in \mathbb{N}$.

Step: assume $n_k \geq k$. Since $n_{k+1} > n_k$ and both are integers, $n_{k+1} \geq n_k + 1 \geq k + 1$. $\square$

Proof of Problem 4. Let $N \in \mathbb{N}$ be given and set $k = N + 1$.

- For any $j \geq k$ we have, by the Lemma, $n_j \geq j$.
- Hence $n_j \geq j \geq k = N + 1 > N$.

So $n_j > N$ for all $j \geq k$, as required. $\square$

Why we care: This says $n_k \rightarrow \infty$, which is precisely what makes the following true:

if $x_n \rightarrow x$ then every subsequence $x_{n_k} \rightarrow x$.

Indeed, given $\varepsilon > 0$ choose N with $\|x_m - x\| < \varepsilon$ for $m \geq N$, then choose k as above: for $j \geq k$, $n_j > N$ and so $\|x_{n_j} - x\| < \varepsilon$.
